

Caleb Bellsmith

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TECHNICAL SKILLS

Languages: C++, C, HTML, LaTeX, Visual Basic, MATLAB, CSS, JavaScript

Tools: SolidWorks, Fusion360, AutoCAD, VSCode, KiCad, CLion, Bamboo, Arduino IDE, GitHub, GitLab

Fabrication: CNC Machining, 3D Printing, Soldering, Manual Prototyping, Woodworking, Welding

EXPERIENCE

Alchemy Nano | *Mechanical Designer and Lab Tech*

2026 – Present

- Performed and optimized lab testing for mechanical and optical properties of Exoshield, an automotive nanocomposite film.
- Designed and manufactured custom testing equipment to enable roll angle evaluation of nanocomposite film samples, expanding in house testing capability.

UWaterloo Space Soldering Team | *Mechanical Designer*

2025 – Present

- Designed and manufactured mechanical systems for experimental centrifuge based microgravity soldering mechanisms.

True North Bio-competition | *Electrical Designer*

2025 – 2026

- Created the electrical control framework for a lower limb exoskeleton through custom PCB design. Used EMG sensors to facilitate intuitive, signal based movement for assistive applications.

Biomechatronics Design Team | *Mechanical Designer*

2025 – 2026

- Developing mechanical components for assistive orthotic devices, focusing on ease of use and ergonomics.

BioTEC Conference | *Sponsorship Lead*

2025 – Present

- Securing partnerships and corporate outreach for Canada's largest student run biotechnology conference.

PROJECTS

EMG Controlled Wireless Mouse

IMU & EMG Sensors, Web Dev

- Created a wearable human machine interface using an ESP32 and IMU to translate arm tilt into mouse movement and EMG sensors to trigger clicks via muscle flexion.
- Developed a full stack web dashboard to monitor device state and calibrate user specific parameters including sensitivity, Bluetooth status, and muscle group mapping.

Precision Robotic Arm

ESP32, C++, Servo Control

- Designed and hand built a custom 5 DoF robotic arm, including circuit design and assembly, high pressure grip systems, and fully 3D printed bearings.
- Currently being integrated with the EMG wireless arm sleeve to enable motion mirroring, translating human arm movement directly into real time robotic actuation.

Custom Cedar Strip Canoe

CAD, CNC Machining

- Computer modeled and fabricated a custom canoe; integrated CNC machined internal molds with traditional woodworking to achieve smooth structural efficiency and aesthetic finish.

Auto Solving Maze

ESP32, C++, Computer Vision

- Developed a servo-powered tilting platform using an ESP32 microcontroller. Integrated a webcam based computer vision system to enable autonomous maze solving and real time pathfinding.

EDUCATION & LEADERSHIP

University of Waterloo | *B.A.Sc. in Biomedical Engineering*

- **WEEF Representative:** Nominated by class to evaluate technical proposals for student project funding and represented class interests.

Lifeguard & Swimming Instructor | *Instructor, NLS, CPR Certified*

2023 – 2025

- Selected for competitive outdoor pool placement (Top 4% of 250+ staff) based on performance and leadership.

Distinctions and Awards

HHCS, TDCH

- Governor General's Academic Medal, Sir Isaac Newton Physics Award, TDCH Valedictorian, HHCS Valedictorian.